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Enhanced Reservoir Definition - A Novel Vectorized Disorder Seismic Attribute for Spatial Incoherency Mapping

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Summary

Although most of the regions in seismic data are usually stratified, coherent and continuous, there are areas in seismic data that look chaotic, i.e., high randomness zones. The randomness of seismic data is mainly due to three reasons:

1. energy of random reflections or refractions from certain subsurface geological structures, e.g. fracture zone, gas chimney, karst collapse, and terminated unconformity;
2. suboptimum acquisition and consequential low data quality, e.g. all kinds of noise;
3. imperfect data processing/imaging procedure, e.g. inaccurate migration velocity, imaging artifacts, and operator aliasing.

Thus, a reliable method to estimate the spatial distribution of the randomness level in seismic data is essential and indispensable for reservoir characterization. Estimation of randomness spatial distribution with high resolution will assist reservoir characterization in multiple ways, including: 1. identify certain geological features such as: fracture zones, gas chimneys, karst collapse, terminated unconformity or other random reflection/refraction related features; 2. indicate seismic data quality and random noise level, e.g. recognizing boundary problem between merged surveys; 3. provide covariance matrix for seismic inversion or uncertainty index for interpretation and reservoir simulation. Thus, randomness level measurement is an important task in seismic image processing/imaging, which helps both to improve seismic processing/imaging quality control (QC) and to better expose some geology structures, such as subsurface anomalies, fracture zones and gas chimneys.

There have been numerous early efforts at estimating the randomness level in seismic data, e.g. cross-correlation-based algorithms, structure-tensor-based algorithms, grey level co-occurrence-matrix-based algorithms, and convolutional filtering-based algorithms. Among them the Disorder algorithm is a convolutional filtering-based algorithm designed using an optimization approach. By design, the Disorder is insensitive to faults, channel edges, unconformities, and other linear discontinuities of geological features.

However, the Disorder algorithm suffers from two major issues: 1. Anisotropy and 2. Non-smoothable. Thus, we proposed novel Vectorized Disorder algorithm, which in comparison to Disorder algorithm provides desirable (much more isotropic and smoother) randomness distribution estimation.

Introduction

Seismic exploration has been the most important approach for hydrocarbon exploration; it uses seismic energy (data) to probe the subsurface structures and locate potential hydrocarbon reservoirs. For finding potential hydrocarbon deposits, geophysical exploration usually uses seismic reflection surveys to image and interpret subsurface structures. In seismic exploration practice, artificial seismic wave generated on the ground travels downward into the earth. The seismic wave bounced back from subsurface interfaces is recorded by a number of seismic receivers and constitute a seismic dataset. The underground structure information can then be extracted from seismic data through seismic imaging and inversion techniques. During seismic data acquisition, the reflection signal is always interfered by random noises from a variety of environment and instrument sources. Furthermore, randomness can also be generated by random reflection or refraction from geological features such as subsurface anomalies, fracture zones, gas chimneys, and the noise is represented by randomness in seismic data/image. Therefore, randomness level measurement is an important tool for seismic data processing/imaging quality control (QC). Furthermore, although most of the regions of seismic data are stratified, coherent and continuous, there are high randomness zones in seismic data that look chaotic. Thus, randomness level measurement, especially randomness level spatial distribution estimation, is an essential task in seismic data processing/imaging.

The randomness of seismic data is mainly due to three reasons:

1. Energy of random reflections or refractions from certain subsurface geological structures, e.g. fraction zone, gas chimney, karst collapse, and terminated unconformity;
2. Suboptimum acquisition and consequential low data quality, e.g. all kinds of noise;
3. Imperfect data processing/imaging procedure, e.g. in- accurate migration velocity, imaging artifacts, operator aliasing.

Thus, a reliable method to estimate the spatial distribution of randomness level in seismic data is essential and indispensable for reservoir characterization. High quality estimation for randomness will assist reservoir characterization in multiple ways, including:

1. Identify certain geological features such as: fracture zones, gas chimneys, karst collapse, terminated unconformity or other random reflection/refraction related features;
2. Indicate seismic data quality and random noise level, e.g. recognizing boundary problem between merged surveys;
3. Provide covariance matrix for seismic inversion or uncertainty index for interpretation and reservoir simulation.

Numerous attempts have been made at estimating the randomness level in seismic data. [Dash and Obaidullah \(1970\)](#) proposed to use cross-correlation to describe noise for seismic traces. Their approach is similar to early coherence algorithms and doesn't work well in faulted areas because faults are accentuated over randomness. [Bahorich and Farmer \(1995\)](#) extended the cross-correlation technique and developed the first coherence attribute to highlight discontinuities in exploration seismic data. [Randen et al. \(2001\)](#) introduced the chaos attribute based on structure tensor, which is an eigen-analysis of a 3x3 gradient components covariance matrix. High chaos values indicate low data directionality. Similar to eigen-coherence, this method is used mainly for fault extraction rather than noise description. The gray level co-occurrence matrix (GLCM) based algorithms are from neural network approach ([West et al., 2002](#); [Gao, 2003](#); [Chopra and Alexeev, 2005](#)). The co-occurrence matrix can distinguish the energy from coherent patterns (lines, curves, blocks, etc.) and from chaotic areas. Based on this matrix, the GLCM entropy is calculated and it can estimate the level of randomness in the data. A large value means higher randomness level in the seismic data and vice versa. In zones near fault planes, seismic data are discontinuous in one direction and continuous in other directions. Faults may also be associated with fracture zones; thus,

local randomness measures may be useful for identifying fracture zones, although this capability has not been tested. All aforementioned methods seem ill suited for distinguishing faults from data randomness. Al-Dossary et al. (2014), Chopra and Marfurt (2016), and Fu and Al-Dossary (2022) proposed Disorder algorithm that differs from its forerunners. The method is based on Immerkaer operator (Immerkaer, 1996) with convolutional filtering method and uses a second order differential operator designed using optimization approach, which is sensitive to chaotic and noisy area but not organized area. By design, the Disorder algorithm is insensitive to faults, channel edges, unconformities, and other linear discontinuities of geological features. Fu and Al-Dossary (2022) demonstrated and analyzed the advantages of Disorder algorithm over all other aforementioned algorithms.

However, the Disorder algorithm suffers from two major issues: 1. An-isotropy and 2. Non-smooth-able. In the following section, we will show where these two flaws of Disorder algorithm come from. The goal of the paper is to propose a novel algorithm to reliably measure the degree of randomness embedded in seismic data/image. The proposed new algorithm is expected to have high capability to measure randomness, and to be highly efficient for parallel execution on modern multi-processor computers/multi-node clusters.

Theory

Vectorized Disorder algorithm is proposed to overcome two major issues of Disorder algorithm while preserving all the advantages of Disorder algorithm. The Disorder algorithm (Al-Dossary et al., 2014; Chopra and Marfurt, 2016; Fu and Al-Dossary, 2022) uses convolutional filtering approach is based Immerkaer operator (Immerkaer, 1996) to remove the energy of regular structures along axial directions in the input data, such as constant values, constant slopes, and steps. After removing energy of regular structures, the energy corresponding to random noises or random signals is extracted from the input data. This algorithm is by far one of the most satisfactory method (Fu and Al-Dossary, 2022) and is widely used in seismic exploration practice. The filter of Disorder algorithm based on an optimization procedure, which removes simple non-random structure such as: constant values, slope, and axial steps. For simplicity, here we only use a 2D case as an example. A 2D Disorder filter will be a 2D convolutional filter. Assuming the randomness energy has a zero mean, this filter will have zero sum of all its coefficients. From the symmetry along axial directions, the coefficients of this filter should be axis symmetric. And since horizontal or vertical constant values, slope, and steps are non-random feature, they should yield zero output of this filter. So, summation of each column or row of the coefficients of this filter is zero, and a 3x3 coefficient-matrix needs to be determined (larger filter size is possible but is usually unnecessary and computational expensive, a 3x3 is a normal size for 2D Disorder filter). Although there are 27 coefficients, only three unique coefficients need to be specified because of the symmetric property mentioned above, while the remaining are predetermined by symmetry.

The three unique to-be-determined coefficients are: (1) a , at the filter center; (2) b , at 4 edge centers; (3) c , at 4 corners. The 2D Disorder filter with to-be-determined coefficients is illustrated in pictorial form as:

$$\begin{bmatrix} c & b & c \\ b & a & b \\ c & b & c \end{bmatrix} \quad (1)$$

The summation of the coefficients from each column or row should be zero, which guarantees to filter out the non-random structures in the input data, such as: constant values, constant slopes, and steps in axial directions. And the coefficient a can be arbitrarily set to 4 to avoid fraction coefficient. This leads to a set of linear equations:

$$\begin{cases} 4 + 2b = 0 \\ b + 2c = 0 \\ 4 + 2c = 0 \end{cases} \quad (2)$$

From linear equations theory, these equations are impossible to solve (except all zeros solution, which is meaningless in our case). The reason is there are only 2 unknowns but we need to satisfy 3 equations (constraints), thus the system is over-determined. To avoid this insolvable over-determined system, the Disorder algorithm has to make a compromise to drop one equation. By assuming the data only contains regular structures along axial directions, the Disorder operator only requires the sum of its coefficients of any axial directions to be 0, but not diagonal direction. Then the system becomes:

$$\begin{cases} 4+2b=0 \\ b+2c=0 \end{cases} \quad (3)$$

This assumption is more or less reasonable in geophysical exploration seismic data/image, given the relatively low probability of appearance of steep structure, but it may be not always true. Solving the equations, the 2D Disorder filter is

$$\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix} \quad (4)$$

Two issues for Disorder

The Disorder algorithm suffers from two major issues:

1. The algorithm is not isotropic, if there are large amount of regular structures along diagonal direction, the Disorder algorithm will not remove them as effectively as in directions parallel to the axis. Therefore, it will provide falsely higher randomness estimation with presence of large amount of diagonal direction features. In other words, Disorder algorithm will produce different randomness distribution estimations when we simply rotate the same input data by 45 degree.
2. The estimated randomness distribution from disorder algorithm cannot be smoothed; in real production we usually want a smooth estimation of distribution for random, but the Disorder algorithm will produce problematic result after smoothing (we will analyze the reason of this phenomenon in the following section.).

It is easy to figure out the first issue. The sums along axial directions are 0, but are not 0 along diagonal directions. The consequence of this anisotropy is that if there is large amount of regular structures along diagonal direction, the Disorder will not be able to remove them and therefor will provide falsely much higher randomness estimation. Also, Disorder algorithm will give different randomness distribution estimations when we simply rotate the input section by 45 degree. But for the second issue, it may not be very obvious. Some derivations might be necessary in spotting the issue.

Disorder is based on convolutional filtering approach, and it is a linear operation:

$$d \circledast f = r \quad (5)$$

where d is input data, $\circledast$ is convolution operation, f is Disorder operator, and r is the randomness distribution estimation result. We want to have a smooth randomness distribution estimation to avoid the impact from a few wild extreme values in data. So we usually smooth the Disorder estimation result r by applying another linear smoothed operator s to get a smoothed randomness distribution r_s :

$$r \circledast s = r_s \quad (6)$$

From linear operator theory, the convolution operation is commutable,

$$(a \circledast b) \circledast c = a \circledast (b \circledast c) \quad (7)$$

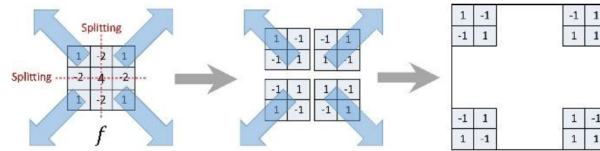
therefore, we have

$$r_s = r \circledast s = (d \circledast f) \circledast s = d \circledast (f \circledast s) \quad (8)$$

The smoothed randomness distribution rs is equal to applying a resultant operator ($f \circledast s$) on the input data d . Without loss of generality, we use a 5-point boxcar smoother for s , the resultant operator, which is the combination of Disorder operator with the smooth operator ($f \circledast s$), is:

$$\begin{array}{ccc}
 \begin{array}{|c|c|c|} \hline 1 & -2 & 1 \\ \hline -2 & 4 & -2 \\ \hline 1 & -2 & 1 \\ \hline \end{array} & \circledast & \begin{array}{|c|c|c|c|c|} \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline 1 & 1 & 1 & 1 & 1 \\ \hline \end{array} \\
 f & & s \\
 \hline
 \begin{array}{|c|c|c|c|c|c|c|c|c|c|} \hline 1 & -1 & 0 & 0 & 0 & 0 & -1 & 1 \\ \hline -1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \hline -1 & 1 & 0 & 0 & 0 & 1 & 1 & -1 \\ \hline 1 & -1 & 0 & 0 & 0 & 1 & 1 & 1 \\ \hline \end{array} \\
 f \circledast s & = & (f \circledast s)
 \end{array}$$

It is obvious that by combining the Disorder and smooth operators, the resultant operator is equal to split the Disorder operator into four pieces and then stretch them into its four corners:



And the consequence of this undesirable resultant operator is that now the input data for the operator are not from the vicinity of the output location anymore, so that the resultant smoothed randomness distribution operator is not local, thus it is not meaningful or useful. That is the fundamental reason why the Disorder algorithm is non-smooth-able.

Vectorized Disorder: fixing the two issues of Disorder

In the previous section we have already shown the two issues of Disorder algorithm and their reasons. The solution of the first flaw is to add extra dimension (degree of freedom) in the operator to eliminate the over-determined nature of the corresponding linear system. And to fix the second flaw, we need to introduce some non-linearity to make the operation for randomness estimation non-linear, which does not commute with the smooth operator.

In this section we propose a new algorithm to estimate randomness: Vectorized Disorder. The new algorithm is based on Vectorized convolution rather than the conventional linear convolution. By exploiting this new technique, we introduce extra dimension (degree of freedom) and non-linearity into the scheme. These two new properties fix both the over-determined equation issue and non-local smoothed operator issue.

Figure 1(a) shows the workflow of Disorder algorithm, which exploits the conventional linear convolution. The conventional linear convolution $\circledast$ is defined as:

$$\begin{aligned}
 & d(x_1, \dots, x_k) \circledast f(x_1, \dots, x_k) \\
 &= \sum_{i_1} \dots \sum_{i_k} d(i_1, \dots, i_k) * f(x_1 - i_1, \dots, x_k - i_k) \\
 &= r(x_1, \dots, x_k)
 \end{aligned} \tag{9}$$

We extend the conventional convolution $\circledast$ by adding one more extra dimension n , and define the vectorized convolution $\circledast_{vec}$ as:

$$\begin{aligned}
& d(x_1, \dots, x_k) \circledast_{vec} f(x_1, \dots, x_k, n) \\
&= \sum_{i_1} \dots \sum_{i_k} d(i_1, \dots, i_k) * f(x_1 - i_1, \dots, x_k - i_k, n) \\
&= r(x_1, \dots, x_k, n)
\end{aligned} \tag{10}$$

Please notice the filter f in the vectorized convolution $\circledast_{vec}$ is not of the same dimension as input data d anymore, it has an extra dimension n , and the output r also has this extra dimension n .

Figure 1(b) shows the workflow of the novel Vectorized Disorder algorithm with vectorized convolution. The output of vectorized convolution has higher dimension than the input data, so we need an extra step after the vectorized convolution to collapse the high dimensional result back into the same dimension as input data, we call this step as reduction. And this reduction is a nonlinear step, and we will show more detail of the reduction step in a later section. After the reduction, the Vectorized Disorder algorithm is not commutable with any further conventional linear convolution afterward. Thus, we solved the non-smoothable issue in the Disorder algorithm.

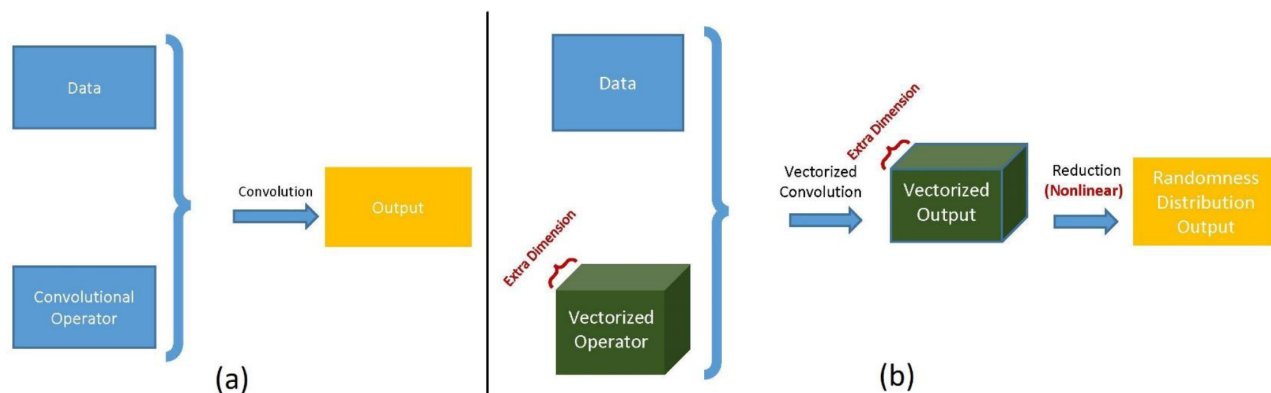


Figure 1—(a) conventional convolution-based Disorder algorithm; (b) novel Vectorized Disorder algorithm, please notice the extra dimension and non-linearity in this new scheme.

The coefficients of the Vectorized Disorder operator

We introduced the extra dimension in the Vectorized Disorder operator. For each component of this extra dimension, we have a complete independent operator in the same dimension as the input data. So, we can assign each of the independent operator specifically to deal with certain regular structures in one single direction. Due to space limitations, here we provide the coefficients of the Vectorized Disorder operator for both 2D and 3D data without derivations. Figure 2(a) shows all the coefficients of 2D Vectorized Disorder operator, and it has 4 components. Figure 2(b) shows all the coefficients of Vectorized Disorder operator in 3D, and it has 13 components, each of them is corresponding to one of 13 unique directions (3 axials, 6 face diagonals, and 4 body diagonals). Each component in the 2D or 3D operator is an individual conventional operator dedicated to structures of one single direction.

$$GR_{2d} = \begin{Bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}_1 \\ \begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}_2 \\ \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & 0 \end{bmatrix}_3 \\ \begin{bmatrix} 0 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{bmatrix}_4 \end{Bmatrix}$$

(a)

$$GR_{3d} = \begin{Bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_1 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_2 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_3 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_4 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_5 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_6 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_7 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_8 \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_9 \end{Bmatrix}$$

(b)

Figure 2—(a) Coefficients of 2D Vectorized Disorder operator, which has 4 components; (b) coefficients of 3D Vectorized Disorder operator, which has 13 components.

The reduction step

The new Vectorized Disorder algorithm is based on vectorized convolution rather than the conventional linear convolution, so the output of the operator has higher dimension (degree of freedom) than the input data. We would like our final result to be of the same dimension as the input data, so we need a reduction step to reduce the extra dimension and provide the result in physical configuration same as the input data. If any component corresponding to certain specific direction has regular structure energy, the randomness of this local area should be low, so we use the minimum function over the extra dimension as the reduction function. The final result s after the reduction is:

$$\begin{aligned} s(x_1, \dots, x_k) &= \min_n (d(x_1, \dots, x_k) \otimes f(x_1, \dots, x_k, n)) \\ &= \min(r(x_1, \dots, x_k, n)) \end{aligned} \quad (11)$$

Parallel implementation on a modern computer/cluster

Furthermore, there is another important benefit of the new Vectorized Disorder algorithm, which is its suitability for parallelization on modern computer/cluster. Taking advantage of vector instructions set (SSE and AVX) of modern Intel and AMD CPUs, or CUDA parallel computing mode on NVidia GPUs, Vectorized Disorder operator is very easy to achieve high efficiency on a modern multi-core computer/multi-node cluster with or without GPU.

Results Comparison

In this section, we compare the results of Disorder and Vectorized Disorder algorithms to demonstrate the advantages of Vectorized Disorder over Disorder. Figure 3 compares the results of Disorder and new Vectorized Disorder algorithms. Please note the Vectorized Disorder algorithm provides much smoother randomness distribution estimation. Furthermore, in the lower-left corner, the Disorder algorithm gives a falsely high randomness measurement, which corresponds to the steep structures in the input data, due to anisotropic nature of Disorder algorithm.

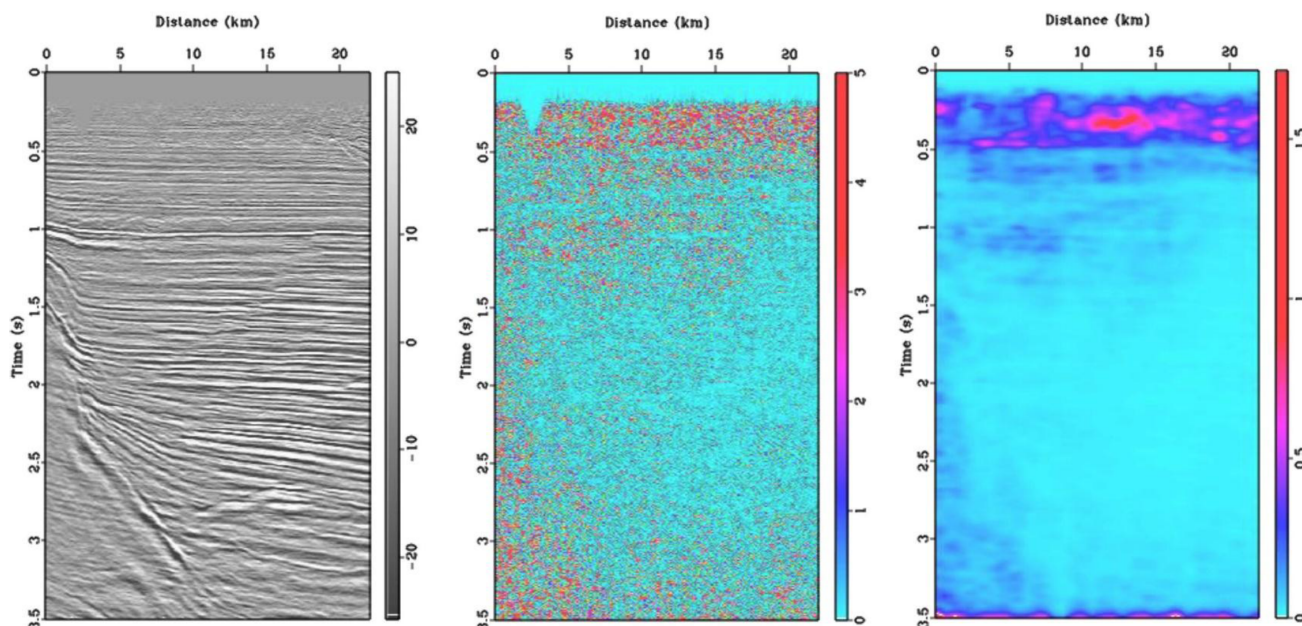


Figure 3—Input seismic data (left); Disorder randomness estimation result (middle); and Vectorized Disorder randomness estimation result (right).

Conclusions

Vectorized Disorder algorithm is proposed to overcome two major issues of Disorder algorithm while preserving all the advantages of Disorder algorithm. It is sensitive to chaotic and noisy data, but is insensitive to faults, channel edges, unconformities, and other linear discontinuities of geological features. It is a powerful innovative seismic algorithm that estimates the spatial distribution of data randomness. The result of Vectorized Disorder is more isotropic and is smooth-able. It can assist reservoir characterization by: 1. identify certain geological features such as fracture zones, gas chimneys, karst collapse, terminated unconformity or other random reflection/refraction related features; 2. indicate seismic data quality and random noise level; 3. provide covariance matrix for seismic inversion or uncertainty index for interpretation and reservoir simulation.

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